

the structure of a G -equivariant bundle – i.e., a G -action on π^*E by vector bundle automorphisms making the projection $\pi^*E \rightarrow X$ equivariant. By fpqc descent, this gives an equivalence of categories

$$\pi^*: \mathbf{Vect}(X/G) \rightarrow \mathbf{Vect}_G(X).$$

That is, we can study vector bundles on X/G in terms of equivariant vector bundles on X . The bundles on X/G which are pulled back along $X/G \rightarrow \text{pt}/G$ correspond to the equivariant vector bundles on X of the form $X \times V \xrightarrow{\pi_1} X$ for some G -representation V , where G acts simultaneously on both factors of $X \times V$. As a special case, vector bundles on $BG = \text{pt}/G$ correspond to equivariant vector bundles on pt , i.e., G -representations.

2.3. Common groups and classes. The following notation is consistent with [Lar21].

Definition 2.1. Throughout this paper, let G denote the wreath product

$$G := (\mathbb{G}_m \times \mathbb{G}_m) \rtimes \mathbb{Z}/2\mathbb{Z},$$

where the action of $\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{G}_m \times \mathbb{G}_m$ permutes its factors.

One can describe representations of G through the sign representation Γ of $\mathbb{Z}/2\mathbb{Z}$ and representations of $\mathbb{G}_m$.

Definition 2.2. We let L_n denote the 1-dimensional representation of $\mathbb{G}_m$ with weight n . Set

$$L_{a_1, \dots, a_n} := L_{a_1} \oplus \dots \oplus L_{a_n}.$$

Let $W_{a_1, \dots, a_n}$ be the representation of G whose underlying vector space is

$$W_{a_1, \dots, a_n} := L_{a_1, \dots, a_n} \oplus L_{a_1, \dots, a_n}$$

and where $\mathbb{G}_m \times \mathbb{G}_m \subset G$ acts naturally on $L_{a_1, \dots, a_n} \oplus L_{a_1, \dots, a_n}$ and $\mathbb{Z}/2\mathbb{Z}$ permutes its factors.

Lastly, we let V denote the standard representation of GL_2 and set $V_n := \text{Sym}^n V^*$ as well as $V_n(m) := V_n \otimes (\det V)^{\otimes m}$.

Definition 2.3. With the above notation, set

$$\begin{aligned} \alpha_i &= c_i(V) \in \text{CH}^*(\text{BGL}_2), \\ \beta_i &= c_i(W_1) \in \text{CH}^*(BG), \\ \gamma &= c_1(\Gamma) \in \text{CH}^*(B\mathbb{Z}/2\mathbb{Z}), \end{aligned}$$

and for $i = 1, 2$ let $t_i \in \text{CH}^*(B(\mathbb{G}_m \times \mathbb{G}_m))$ be the pullback of $c_1(L_1)$ along the i -th projection map $B(\mathbb{G}_m \times \mathbb{G}_m) \rightarrow B\mathbb{G}_m$.

Whenever we are working in the Chow ring of a space equipped with a map to BGL_2 , BG , or $B(\mathbb{G}_m \times \mathbb{G}_m)$, we will denote by α_i , β_i , etc., the pullbacks of these classes from $\text{CH}^*(\text{BGL}_2)$, etc. We have the following explicit presentations of these rings:

Lemma 2.4 ([Tot14, Theorem 2.13 and Lemma 2.12; Lar21, Theorem 5.2 and Lemma 7.1]). *We have the presentations*

$$\text{CH}^*(\text{BGL}_2) \cong \mathbb{Z}[\alpha_1, \alpha_2], \quad \text{CH}^*(BG) \cong \mathbb{Z}[\beta_1, \beta_2, \gamma]/(2\gamma, \gamma^2 + \beta_1\gamma), \quad \text{CH}^*(B(\mathbb{G}_m \times \mathbb{G}_m)) \cong \mathbb{Z}[t_1, t_2].$$

Moreover, the norm (or transfer map) from $\mathbb{G}_m \times \mathbb{G}_m$ to G – in other words, the pushforward $\pi_: \text{CH}^*(\mathbb{G}_m \times \mathbb{G}_m) \rightarrow \text{CH}^*(G)$ along the index 2 inclusion $\mathbb{G}_m \times \mathbb{G}_m \hookrightarrow G$ – is given by*

$$\begin{aligned} \pi_*(1) &= 2 & \pi_*(t_1^a) &= \beta_1 \pi_*(t_1^{a-1}) - \beta_2 \pi_*(t_1^{a-2}) \text{ for } a \geq 2 \\ \pi_*(t_1) &= \beta_1 + \gamma & \pi_*(t_1^a t_2^b) &= \beta_2^{\min(a,b)} \pi_*(t_1^{|a-b|}), \end{aligned}$$